

PARAMETRIC

Assume that data follows a specific distribution and are characterized by a fixed set of parameters, eg. $N(\mu, \sigma^2)$.

Function:

Model and analyse relationship between one dependent variable and one or more independent variables to understand associations.

Topics:

- Simple linear regression
 $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$
- Multiple linear regression
 $y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i$
- Logistic regression
 $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i$
- Model building

Function:

Method for analysing data collected over time to identify patterns, trends, seasonality, and to make forecasts or understand underlying processes.

Topics:

- Time series components
- Moving average smoothing
- Classical decomposition: $Y_t = T_t \times S_t \times R_t$
- Forecaster's toolbox
 - Simple forecasting methods: average, naïve, seasonal naïve, drift
 - Transformations
 - Check residuals
 - Exponential smoothing: SES, Holt, Holt-Winters
 - Autocorrelation & partial autocorrelation plots
 - ARIMA: $X_t = c + \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$

$AR(p)$
 $MA(q)$

(ALMOST) EVERYTHING IS RELATED TO STATISTICAL MODELLING

Regression

AN

- linear models
- different purpose: relationship vs mean differences
- partitions variance (explained and unexplained components)
- use F-tests
- R: contin. predictors
- AN: categorical groups

TS

- can be used for prediction
- finds estimates for model parameters
- TS can include R components
- models relationship between variables
- TS: autocorrelated obs.
- R: independent obs.
- TS: model patterns + behaviours over time
- R: Explain relationships between variables

NP

- test for specific hypotheses
- assess relationships or associations
- NP used when R assumptions (normality or linearity) not met.
- R: specific model for NP: no distributional assumptions
- R: interval/ratio scale data
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Time Series

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ANOVA

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Non-parametrics

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NON-PARAMETRIC

No strong assumption is made about the data distribution and these methods rely on fewer/no parameters (e.g. ranks or medians).

Function:

Test whether there are significant differences between the means of three or more groups.

Topics:

- Experimental Design
- Completely Randomised Design
 $Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$
- Randomised Complete Block Design
 $Y_{ijk} = \mu + \alpha_i + \beta_j + \epsilon_{ijk}$
- Factorial Experiments
 $Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + \epsilon_{ijk}$

Function:

Methods that don't assume specific distributions, often based on ranks or medians: suitable for non-normal or ordinal data.

Topics:

- Data types
- Wilcoxon signed rank sum test (2 dep. samples)
- Mann-Whitney U-test (2 indep. samples)
- Kruskal-Wallis test (3+ indep. samples)
- Friedman test (3+ dep. samples)
- Spearman's rank correlation test

S = similarities
D = differences